Concentration New Course

CS156: Finding Patterns in Data with Machine Learning

Term	Credit
Fall 2026	4

Course Description

Students learn to apply core machine learning techniques — such as classification, perceptron, neural networks, support vector machines, hidden Markov models, and nonparametric models of clustering — as well as fundamental concepts such as feature selection, cross-validation and over-fitting. Students program machine learning algorithms to make sense of a wide range of data, including images and numerical data.

Course Objectives & Learning Outcomes

Core machine learning knowledge and skill mastery.

#cs156-MLCode: Produce working, readable, and performant Python implementations of a variety of machine learning systems using appropriate libraries and software tools.

#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

#cs156-MLMath: Evaluate problems and derive solutions in linear algebra, multivariate calculus, and Bayesian statistics.

Produce original machine learning work that goes beyond the scope of what was covered in class.

#cs156-MLDevelopment: Contribute to the quality of ML resources for current and future students by compiling learning resources, sharing code, creating study groups, and supporting other student's learning.

#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Prerequisites & Working Knowledge

CS110 Problem Solving with Data Structures and Algorithms

CS111 Single and Multivariable Calculas

CS113 Theory and Applications of Linear Algebra

CS114 Probability, Statistics, and the Structure of Randomness

Assignments

Note: Sunday is considered the beginning of the academic week for determining due dates.

Assignment Title	Weight	Due	Released	Closed
Repository and Summary	12x	Fri, Week 1	Mon of Week 1	Mon of Week 2
First Pipeline	4x	Thu, Week 6	Mon of Week 1	Sun of Week 7
Second Pipeline	6x	Mon, Week 13	Mon of Week 7	Thu of Week 13
Final Pipeline	8x	Thu, Week 15	Mon of Week 12	Fri of Week 15

Course Scoring

Type	% of Course Score
	75%

Assignments	
First Pipeline	10%
Repository and Summary	30%
Second Pipeline	15%
Final Pipeline	20%

Classroom Scores	**25%**
Pre-Class Work	15%
Poll Responses & Makeup Work	10%
Verbal Contributions	0%
Breakouts	0%

Required Texts

(Main Text) Murphy, K. P. (2022) Probabilistic Machine Learning (Creative Commons)

https://probml.github.io/pml-book/book1.html

(Supplemental) Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press. (Free online Access)

http://web4.cs.ucl.ac.uk/staff/D.Barber/pmwiki/pmwiki.php?n=Brml.HomePage

(Supplemental) Bishop, C. M. (2006). *Pattern recognition and machine learning*. New York: Springer. (Free online Access) https://www.microsoft.com/en-us/research/people/cmbishop/prml-book/

https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf

(Supplemental) Downey, A. (2013). *Think Bayes*. Beijing: O'Reilly. (Creative Commons)

http://www.greenteapress.com/thinkbayes/thinkbayes.pdf

(Supplemental) MacKay, D. J. (2003). *Information theory, inference, and learning algorithms*. Cambridge, UK: Cambridge University Press. (Free online Access)

http://www.inference.phy.cam.ac.uk/itila/book.html

Schedule of Topics & Readings

This course meets for 2 class sessions each week.

Unit 0: ML Fundamentals

Machine learning stands at the intersection of statistics and computer science. It is a greatly promising set of ideas that allow us to extract valuable information from the large volume of data that society is currently producing. In this unit we provide a basic framework we can use to begin to answer some of the more basic questions: What is a machine-learning algorithm? What data are appropriate for a given algorithm? How should we prepare the data and then validate the model afterwards? If we have a model, how do we decide on any actions to take based on the model's output?

Session 1: Introduction and overview of machine learning concepts

Learning Outcomes

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#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

Reading, Videos, and Other Preparation Resources

Read Chapter 1 of Murphy, K (2022). *Probabilistic Machine Learning*

Machine Learning: Solving Problems Big, Small, and Prickly. (2016, December 12). Retrieved August 16, 2017, from https://www.youtube.com/watch?v=_rdINNHLYaQ

https://www.youtube.com/watch?v=_rdINNHLYaQ

Video Intro to CS156: Machine Learning

https://youtu.be/WN7rCCloLbQ

Video Machine Learning-Four Core Skills

https://youtu.be/36ntbsE8n74

Machine Learning Motivating Examples

https://youtu.be/Gb8Ex7boHrk

What Actually is a Machine Learning Model?

https://youtu.be/crkz2vLn9nw

Booting up--quick class instructions for the first session.

https://youtu.be/RwURUy-Whxo

Class Learning Evaluation and Major Assignments

https://youtu.be/ej7bVPto-kY

Unit 1: Core Mathematics

From the other side of the screen it all looks so easy.
— Kevin Flynn
This unit reviews linear models and statistical techniques from previous classes in the context of scikit learn. It introduces ML data ingestion and engineering techniques, and some simple models for classification, and regression. As well as some unsupervised techniques that capture data structure including PCA and maximum likelihood learning.

Session 2: Linear Algebra 1: Tensors, Classification, and Regression

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#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Basic Linear Regression using the Iris Data set.

https://medium.com/analytics-vidhya/linear-regression-using-iris-dataset-hello-world-of-machine-learning-b0feecac9cc1

Basic Logistic Regression using the Iris Data set.

https://scikit-learn.org/stable/auto_examples/linear_model/plot_iris_logistic.html

Stat Quest: Linear Regresstion

https://www.youtube.com/watch?v=nk2CQITm_eo

Stat Quest: Logistic Regression

https://www.youtube.com/watch?v=yIYKR4sgzI8

[Extension] Read the "Linear regression" and "Logistic regression" Sections in (Main Text) Murphy, K. P. (2022) Probabilistic Machine Learning (Creative Commons)

https://probml.github.io/pml-book/book1.html

Session 3: Linear Algebra 2: Collinearity

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Reading, Videos, and Other Preparation Resources

Murphy, K (2022).

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

The Essence of Linear Algebra (3Blue1Brown, 2016)

https://www.youtube.com/watch?v=kjBOesZCoqc&list=PL0-GT3co4r2y2YErbmuJw2L5tW4Ew2O5B

Multiple regression, clearly explained (StatQuest, Josh Starmer, 2017)

https://www.youtube.com/watch?v=zITIFTsivN8&list=PLblh5JKOoLUIzaEkCLIUxQFjPIlapw8nU&index=4

Session 4: Trees 1: Classification by Partition

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#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Starmer, J. (2021, April 26). *Decision and Classification Trees, Clearly Explained!!!*

https://www.youtube.com/watch?v=_L39rN6gz7Y

Wilber, J., & Santamaría, L. (n.d.). *Decision Trees*. Retrieved October 14, 2022

https://mlu-explain.github.io/decision-tree/

scikit-learn developers. (n.d.). *1.10. Decision Trees*. Scikit-Learn. Retrieved October 14, 2022

https://scikit-learn/stable/modules/tree.html

Murphy, K. P. (2022) Probabilistic Machine Learning. Chapter 18.1

https://probml.github.io/pml-book/book1.html

Session 5: Max Likelihood 1: Naive Bayes

Learning Outcomes

Reading, Videos, and Other Preparation Resources

Starmer, J. (2020). *Naive Bayes, Clearly Explained!!!*

https://youtu.be/O2L2Uv9pdDA

Starmer, J. (2020). *Gaussian Naive Bayes, Clearly Explained!!!*

https://www.youtube.com/watch?v=H3EjCKtlVog

VanderPlas, J. (2016). *In Depth: Naive Bayes Classification*

https://jakevdp.github.io/PythonDataScienceHandbook/05.05-naive-bayes.html

Session 6: Max likelihood 2: Parameter Estimation from Data

Learning Outcomes

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Reading, Videos, and Other Preparation Resources

A Probability is not Likelihood (StatQuest, Starmer, J 2018).

https://www.youtube.com/watch?v=pYxNSUDSFH4

Maximum Likelihood (StatQuest, Starmer, J. 2021)

https://www.youtube.com/watch?v=XepXtl9YKwc

Neural Language Models (Shafkat I., 2022 Nov)

https://irhum.github.io/blog/lm/#maximum-likelihood-estimation

Maximum Likelihood for Logistic Regression (Ritvik Kharkar, 2019)

https://www.youtube.com/watch?v=VOIhswqFWVc

Skim Chapter 4 Murphy, K (2022), sections 4.1, 4.2 and 4.3.

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

[Optional] Here are some additional maximum Likelihood Estimation Problems for practice.

https://www.probabilitycourse.com/chapter8/8_2_3_max_likelihood_estimation.php

Session 7: Networks 1: Feed-Forward Neural Networks

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Reading, Videos, and Other Preparation Resources

Murphy, K (2022)
Chapter 13:
13.0 (Introduction)
13.1
13.2
13.3.1
13.3.2

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

But what is a neural network? (3Blue1Brown 2017)

https://www.youtube.com/watch?v=aircAruvnKk

Gradient descent (3Blue1Brown 2017)

https://www.youtube.com/watch?v=IHZwWFHWa-w&t=456s

Session 8: Gradients: Multivariate Derivatives

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#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

CS231n. (n.d.). *Backpropagation, Intuitions*. Retrieved September 30, 2022.

https://cs231n.github.io/optimization-2/

Smilkov, D., & Carter, S. (n.d.). *A Neural Network Playground*. Retrieved September 30, 2022,

https://playground.tensorflow.org/

[Review] Murphy, K. P. (2022) Probabilistic Machine Learning. Chapters 13.3.1 and 13.3.2 (from Session 9) and Chapter 13.3.4 (new)

https://probml.github.io/pml-book/book1.html

[Optional] Shafkat, I. (2021). *Intro to Machine Learning with JAX*.

https://www.youtube.com/watch?v=H_lfVUh03cM

[Optional] Jacobians, RitvikMath, 2021

https://www.youtube.com/watch?v=JsPFlmewq54

Session 9: Metrics and Cross-Validation

Learning Outcomes

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#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Wilber, J. & Werness, B. (n.d.). *The Importance of Data Splitting. MLU-Explain*

https://mlu-explain.github.io/train-test-validation/

Wilber, J. (2022, March). *Precision & Recall. MLU-Explain*

https://mlu-explain.github.io/precision-recall/

Wilber, J. (2022, June). *ROC & AUC. MLU-Explain*

https://mlu-explain.github.io/roc-auc/

Wilber, J. & Croome, J. (2022, May). *Cross Validation. MLU-Explain*

https://mlu-explain.github.io/cross-validation/

A Gentle Introduction to k-fold cross validation (J. Brownlee 2019)

https://machinelearningmastery.com/k-fold-cross-validation/

Cross-validation sklearn documentation

https://scikit-learn.org/stable/modules/cross_validation.html

[Optional] Murphy, K (2022) Sections 4.5.1 and 4.5.5

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

Session 10: Review 1: Unit 1 Synthesis and Assignment 1 Prep

Learning Outcomes

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#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Unit 2: High-dimensional Data

"Ignorance is bliss"
– Cypher
This unit introduces high-dimensional data and the challenges this introduces for simplistic ML approaches. It begins by characterizing overfitting, cross-validation, and the bias-variance trade off. It continues with the challenges of high-dimensional spaces and how to use linear algebra to estimate values within them. The unit introduces a number of more complex ML models including SVMs, CNNs, Random forests, and Autoencoders.

Session 11: Data Ethics and the Bias-Variance Trade off

Learning Outcomes

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Reading, Videos, and Other Preparation Resources

Mehotra, K. (2022, July 23). *Human Touch*. Fifty Two.

https://fiftytwo.in/story/human-touch/

Wilber, J. & Werness, B. (2021, January). *The Bias Variance Tradeoff. MLU-Explain*

https://mlu-explain.github.io/bias-variance/

Bias-Variance, Kilian Weinberger Cornell Open Lectures.

https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html

Federal Data Strategy Data Ethics Framework (2019)

https://resources.data.gov/assets/documents/fds-data-ethics-framework.pdf

[Optional, Alternate A] Bias-Variance Murphy, K (2022), Chapter 4 section 4.7.6.

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

[Optional, Alternate B] Bias-Variance Trade off, Wikipedia.

https://en.wikipedia.org/wiki/Bias%E2%80%93variance_tradeoff

[Optional but interesting] Machine learning techniques for biased data (MIT, 2021)

https://news.mit.edu/2022/machine-learning-biased-data-0221

Session 12: Basis Functions 1: Functions as Parameters

Learning Outcomes

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#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

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Reading, Videos, and Other Preparation Resources

Murphy, K (2022)
Chapter 12: General Linear Models (short)

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

Linear Regression and Basis Functions (2017)

https://www.youtube.com/watch?v=rVviNyIR-fI

Basis Functions (Starke, P, 2019)

https://www.youtube.com/watch?v=xTo8Q9asVfc

Session 13: Projections 1: Support Vector Machines & Kernels

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

#cs156-MLCode: Produce working, readable, and performant Python implementations of a variety of machine learning systems using appropriate libraries and software tools.

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#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Rohrer, B. (2017, September 17). How Support Vector Machines work / How to open a black box [video file].

https://www.youtube.com/watch?v=-Z4aojJ-pdg

[3Blue1Brown]. (2016, August 15). Inverse matrices, column space and null space | Essence of linear algebra, chapter 6 [video file].

https://www.youtube.com/watch?v=uQhTuRlWMxw

Chapter 14.5
Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective.* Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

SVM Kernels (Ritvikmath, 2019)

https://www.youtube.com/watch?v=OKFMZQyDROI

Ranjan, C. (2017, July 26). Data Science for Mortals - The Kernal Trick [Blog post].

https://dscm.quora.com/The-Kernel-Trick

Session 14: Trees 2: XGBoost

Learning Outcomes

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Reading, Videos, and Other Preparation Resources

xgboost developers. (n.d.). *Introduction to Boosted Trees—Xgboost 1.6.2 documentation*.

https://xgboost.readthedocs.io/en/stable/tutorials/model.html

Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

https://arxiv.org/abs/1603.02754

[Optional] Murphy, K. P. (2022) Probabilistic Machine Learning. Chapter 18.5.5

https://probml.github.io/pml-book/book1.html

Session 15: Basis Functions 2: Ridge and Lasso

Learning Outcomes

#optimization: Evaluate and apply optimization techniques appropriately. (C) FA

#regression: Apply and interpret regression. (C) FA

Reading, Videos, and Other Preparation Resources

Starmer, J. (2018, September 25)._ Regularization Part 1: Ridge (L2) Regression._

https://www.youtube.com/watch?v=Q81RR3yKn30

Starmer, J. (2018, October 1)._ Regularization Part 2: Lasso (L1) Regression._

https://www.youtube.com/watch?v=NGf0voTMlcs

Starmer, J. (2020, May 19)._ Ridge vs Lasso Regression, Visualized!!!._

https://www.youtube.com/watch?v=Xm2C_gTAl8c

[Review] Wilber, J. & Werness, B. (2021, January). *The Bias Variance Tradeoff. MLU-Explain*

https://mlu-explain.github.io/bias-variance/

Parr, T. (2020, May) *A visual explanation for regularization of linear models.*

https://explained.ai/regularization/index.html

Session 16: Dimensionality reduction 1: PCA

Learning Outcomes

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Reading, Videos, and Other Preparation Resources

Starmer, J. (2018). *StatQuest: Principal Component Analysis (PCA), Step-by-Step*.

https://www.youtube.com/watch?v=FgakZw6K1QQ

Powell, V. & Lehe, L. (2009). *Principal Component Analysis: Explained Visually.*

https://setosa.io/ev/principal-component-analysis/

Richardson, M. (2009). *Principal Component Analysis*

https://people.duke.edu/~hpgavin/SystemID/References/Richardson-PCA-2009.pdf

VanderPlas, J. (2016). *In Depth: Principal Component Analysis*

https://jakevdp.github.io/PythonDataScienceHandbook/05.09-principal-component-analysis.html

[Optional, strongly recommended] Macken, F. (2022) *CS156 4.1 (Principal Component Analysis)*

https://www.notion.so/CS156-4-1-Principal-Component-Analysis-98ece2a6218c4f23aa08d1bf9aaa3cec

[Review] 3Blue1Brown: The essence of Linear Algebra (2016), Chapter 13: Change of Basis

https://www.youtube.com/watch?v=P2LTAUO1TdA

[Review] 3Blue1Brown: The essence of Linear Algebra (2016), Chapter 14: Eigenvectors

https://www.youtube.com/watch?v=PFDu9oVAE-g&t=182s

[Optional] Sections 20.1 and 20.2 of Murphy, K (2022)

https://github.com/probml/pml-book/releases/latest/download/book1.pdf

Session 17: Projections 2: Filters, Convolutions, & Transforms

Learning Outcomes

Reading, Videos, and Other Preparation Resources

Shafkat, I. (2018, June 1). *Intuitively Understanding Convolutions for Deep Learning*. Towards Data Science.

https://towardsdatascience.com/intuitively-understanding-convolutions-for-deep-learning-1f6f42faee1

Olah, C. (2014, July 8). *Conv Nets: A Modular Perspective—Colah’s blog*.

https://colah.github.io/posts/2014-07-Conv-Nets-Modular/

Olah, C., Mordvintsev, A., & Schubert, L. (2017). Feature Visualization. *Distill*.

https://doi.org/10.23915/distill.00007

[Optional] Murphy, K. P. (2022) Probabilistic Machine Learning. Chapters 14.1 and 14.2.1

https://probml.github.io/pml-book/book1.html

Session 18: Dimensionality Reduction 2: Autoencoders

Learning Outcomes

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Reading, Videos, and Other Preparation Resources

Jordan, J. (2018, March 19). *Introduction to autoencoders.* Jeremy Jordan.

https://www.jeremyjordan.me/autoencoders/

Artificial Intelligence - All in One. (2017, September 24). *Lecture 15.1—From PCA to autoencoders—[Deep Learning* | *Geoffrey Hinton* | *UofT]*.

https://www.youtube.com/watch?v=PSOt7u8u23w

Artificial Intelligence - All in One. (2017, September 24). *Lecture 15.2—Deep autoencoders—[Deep Learning* | *Geoffrey Hinton* | *UofT]*.

https://www.youtube.com/watch?v=6jhhIPdgkp0

Building Autoencoders in Keras. (n.d.). Retrieved October 21, 2022

https://blog.keras.io/building-autoencoders-in-keras.html

[Optional] Murphy, K. P. (2022) Probabilistic Machine Learning. Chapters 20.3.1 and 20.3.2

https://probml.github.io/pml-book/book1.html

Session 19: Time Series 2: Recurrent neural networks

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

Karpathy, A. (2015, May 21). *The Unreasonable Effectiveness of Recurrent Neural Networks*.

http://karpathy.github.io/2015/05/21/rnn-effectiveness/

StatQuest with Josh Starmer. (2022, July 11). *Recurrent Neural Networks (RNNs), Clearly Explained!!!*

https://www.youtube.com/watch?v=AsNTP8Kwu80

[Optional] StatQuest with Josh Starmer. (2022, November 7). *Long Short-Term Memory (LSTM), Clearly Explained*.

https://www.youtube.com/watch?v=YCzL96nL7j0

[Optional] Murphy, K. P. (2022) Probabilistic Machine Learning. Chapter 15.2

https://probml.github.io/pml-book/book1.html

Session 20: Networks 2: Transfer Learning

Learning Outcomes

Reading, Videos, and Other Preparation Resources

Ruder, S. (2017) *Transfer Learning - Machine Learning's Next Frontier*

https://www.ruder.io/transfer-learning/

Olah, C. (2015) *Visualizing Representations: Deep Learning and Human Beings*

https://colah.github.io/posts/2015-01-Visualizing-Representations/

Ruder, S. (2019) *The State of Transfer Learning in NLP*

https://www.ruder.io/state-of-transfer-learning-in-nlp/

[Skim] Howard, J. & Ruder, S. (2018). *Universal Language Model Fine-tuning for Text Classification*

🔗 https://arxiv.org/abs/1801.06146

[Skim] Brown et. al. (2020) *Language Models are Few-Shot Learners*

🔗 https://arxiv.org/abs/2005.14165

[Optional] Radford et. al. (2018) *Improving Language Understanding by Generative Pre-Training*

🔗 https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

[Optional] Radford. et. al. (2019) *Language Models are Unsupervised Multitask Learners*

🔗 https://d4mucfpksywv.cloudfront.net/better-language-models/language-models.pdf

Session 21: Review 2: Unit 2 Synthesis and Assignment 2 Prep

Learning Outcomes

#cs156-MLCode: Produce working, readable, and performant Python implementations of a variety of machine learning systems using appropriate libraries and software tools.

#cs156-MLDevelopment: Contribute to the quality of ML resources for current and future students by compiling learning resources, sharing code, creating study groups, and supporting other student's learning.

#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Unit 3: Generative Models

- Del Spooner: Human beings have dreams. Even dogs have dreams, but not you, you are just a machine. An imitation of life. Can a robot write a symphony? Can a robot turn a canvas into a beautiful masterpiece?

- Sonny: Can you?

Session 22: Time Series 1: Hidden Markov models

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

Normalized Nerd (2020) *Markov Chains Clearly Explained! Part - 1*

https://youtu.be/i3AkTO9HLXo

Normalized Nerd (2020) *Hidden Markov Model Clearly Explained! Part - 5*

https://youtu.be/RWkHJnFj5rY

Yoon B. J. (2009). *Hidden Markov Models and their Applications in Biological Sequence Analysis*.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2766791/

Jurafsky D. & Martin J. H. (2023). *Hidden Markov Models.*

https://web.stanford.edu/~jurafsky/slp3/A.pdf

Session 23: Attention

Learning Outcomes

Reading, Videos, and Other Preparation Resources

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems*, *30*.

https://papers.nips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

What is a Transformer [3b1b]

🔗 https://www.youtube.com/watch?v=wjZofJX0v4M

Attention in Transformers [3b1b]

🔗 https://www.youtube.com/watch?v=eMlx5fFNoYc&list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi&index=7&t=5s

Session 24: Transformer-Based Large Language Models

Learning Outcomes

Reading, Videos, and Other Preparation Resources

Hill, F. (2023). *Why transformers are obviously good models of language.*

🔗 https://drive.google.com/file/d/1Z4raW0RRzGIvHeXtCRb0xjRil_4KBkWE/view

Raschka, S. (2023). *Understanding and Coding the Self-Attention Mechanism of Large Language Models From Scratch.*

🔗 https://sebastianraschka.com/blog/2023/self-attention-from-scratch.html

Shafkat, I. (2022). *Neural Language Models.*

🔗 https://irhum.github.io/blog/lm/#part-b-language-models

Gwern (2022). *The Scaling Hypothesis.*

🔗 https://gwern.net/scaling-hypothesis#why-does-pretraining-work

Session 25: Review 3: Course Synthesis and Final Project Prep

Learning Outcomes

#breakitdown: Organize problems into tractable components and design solutions. (H) EA

#cs156-MLCode: Produce working, readable, and performant Python implementations of a variety of machine learning systems using appropriate libraries and software tools.

#cs156-MLDevelopment: Contribute to the quality of ML resources for current and future students by compiling learning resources, sharing code, creating study groups, and supporting other student's learning.

#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

How go give a lightning talk (Woolston 2021)

https://www.nature.com/articles/d41586-021-01674-9

Unit 8: Alternate Lessons

Session 28: Fisher’s Linear discriminant

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

(Option 1) Read Chapter 16 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

(Option 2) Read section 8.6.3 of Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective*. Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

Session 28: Gaussian processes

Learning Outcomes

#regression: Apply and interpret regression. (C) FA

Reading, Videos, and Other Preparation Resources

De Freitas, N. (2013, February 04). Introduction to Gaussian processes. Retrieved September 22, 2017, from https://www.youtube.com/watch?v=4vGiHC35j9s

https://www.youtube.com/watch?v=4vGiHC35j9s

(Optional) Read sections 19.1-19.4 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

Bailey, K. *Gaussian Processes for Dummies*. Retrieved March 03, 2019, from http://katbailey.github.io/post/gaussian-processes-for-dummies/

http://katbailey.github.io/post/gaussian-processes-for-dummies/

(Optional) Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective*. Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

(Optional) Cunningham, J. (2012, July 06). MLSS 2012: J. Cunningham - *Gaussian Processes for Machine Learning (Part 1)*. Retrieved March 03, 2019, from https://www.youtube.com/watch?v=BS4Wd5rwNwE

https://www.youtube.com/watch?v=BS4Wd5rwNwE

Session 29: Topic modeling

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

Chen, E. (n.d.). Introduction to Latent Dirichlet Allocation. Retrieved October 05, 2017, from http://blog.echen.me/2011/08/22/introduction-to-latent-dirichlet-allocation/

http://blog.echen.me/2011/08/22/introduction-to-latent-dirichlet-allocation/

(Optional) Blei, D.; Ng, A.; Jordan, M (2003). "Latent Dirichlet Allocation". *Journal of Machine Learning Research*. **3** (4–5): *pp.* 993–1022.

https://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf

(Optional) Read Sections 20.6 of Barber, D. (2012).*Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

(Optional) Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective.* Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

Session 29: Bayesian decision theory

Learning Outcomes

#probability: Apply and interpret fundamental concepts of probability, including conditional and Bayesian probabilities. (C) FA

#utility: Use utility to represent preferences and analyze purposeful decision-making. (C) FA

#gametheory: Apply and evaluate game-theoretic models. (C) FA

Reading, Videos, and Other Preparation Resources

(Option 1) Read sections 7.1 and 7.2 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

Chorionic villus sampling (Beyond the Basics). (n.d.). Retrieved August 17, 2017, from http://www.uptodate.com/contents/chorionic-villus-sampling-beyond-the-basics

http://www.uptodate.com/contents/chorionic-villus-sampling-beyond-the-basics

(Option 2) Murphy, K. (2012). _Machine Learning: A Probabilistic Perspective. _Cambridge: The MIT Press.

- 2.2.3.1 Example: medical diagnosis (p.29): read through the example as it may help with pre-class work
- 5.7.3.2 Utility theory (p.185): provides a short introduction to utility theory
- 10.6 Influence (decision) diagrams (p.328-331): read the graphical interpretation of decision diagrams up to POMDP.

https://probml.github.io/pml-book/book0.html

Session 32: Expectation maximization

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

(Option 1) Read Sections 11.1 - 11.4 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

Do, C. B., & Batzoglou, S. (2008). What is the expectation maximization algorithm?. *Nature biotechnology*, *26*(8), 897. Reading found in EBSCO Academic Search Complete database

(Optional) Chapter 9 of Bishop, C. M. (2006). *Pattern recognition and machine learning*. New York: Springer.

(Option 2) Read section 11.4 of Murphy, K. (2012). Machine Learning: A Probabilistic Perspective. Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

(Optional) Iacobelli F (2015). The EM Algorithm. Retrieved from Youtube on 4 November 2019. https://www.youtube.com/watch?v=7e65vXZEv5Q

https://www.youtube.com/watch?v=7e65vXZEv5Q

Session 33: Bayesian linear models

Learning Outcomes

#probability: Apply and interpret fundamental concepts of probability, including conditional and Bayesian probabilities. (C) FA

#regression: Apply and interpret regression. (C) FA

Reading, Videos, and Other Preparation Resources

(Option 1) Read section 18.1 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

(Optional) Mathematicalmonk (2011, June 29). Bayesian Linear Regression. Retrieved August 29, 2017, from https://www.youtube.com/watch?v=dtkGq9tdYcI

https://www.youtube.com/watch?v=dtkGq9tdYcI

(Option 2) Murphy, K. (2012). Machine Learning: A Probabilistic Perspective. Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

Brooks-Bartlett, J. (2018) *Probability concepts explained: Bayesian inference for parameter estimation.*

https://towardsdatascience.com/probability-concepts-explained-bayesian-inference-for-parameter-estimation-90e8930e5348

Koehrsen, W. (2018) *Introduction to Bayesian Linear Regression.*

https://towardsdatascience.com/introduction-to-bayesian-linear-regression-e66e60791ea7

Session 34: Dimensionality Reduction 3: Gaussian Mixture Models

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

Bekkers, E. (2020, November 27) *9.1 Unsupervised Learning: Latent Variable Models (UvA - Machine Learning 1 - 2020)*

https://www.youtube.com/watch?v=zFHwYvmDCCg

Bekkers, E. (2020, November 27) *9.4 Gaussian Mixture Models And Expectation Maximization (UvA - Machine Learning 1 - 2020)*

https://www.youtube.com/watch?v=TfMtshd5mgo

CS228 (n.d.) *Learning in latent variable models*

https://ermongroup.github.io/cs228-notes/learning/latent/

VanderPlas, J (2016) *In Depth: Gaussian Mixture Models*

https://jakevdp.github.io/PythonDataScienceHandbook/05.12-gaussian-mixtures.html

Session 35: Networks 3: Distributions as Bayesian Networks

Learning Outcomes

#dataviz: Interpret, analyze, and create data visualizations. (C) EA

#probability: Apply and interpret fundamental concepts of probability, including conditional and Bayesian probabilities. (C) FA

Reading, Videos, and Other Preparation Resources

[Review] Bayes Theorem (3b1b 2018)

https://www.youtube.com/watch?v=HZGCoVF3YvM

Bayesian networks and inference over graphs (Sanford 2019)

https://www.youtube.com/watch?v=U23yuPEACG0

[Optional In depth reading for Extension] Read sections 8.1 and 8.2 of
Bishop, C. M. (2006). *Pattern recognition and machine learning*. New York: Springer.

https://www.microsoft.com/en-us/research/wp-content/uploads/2016/05/Bishop-PRML-sample.pdf

Session 40: Quantifying model quality

Learning Outcomes

#dataviz: Interpret, analyze, and create data visualizations. (C) EA

#modeling: Recognize how models can be used to describe a system, explain a set of data, and generate predictions. (C) EA

#regression: Apply and interpret regression. (C) FA

Reading, Videos, and Other Preparation Resources

Read wikipedia's article on Sensitivity and Specificity.

https://en.wikipedia.org/wiki/Sensitivity_and_specificity

Scikit learn documentation, Chapter 3. Model selection and evaluation. (2010). Retrieved February 23, 2017, from http://scikit-learn.org/stable/model_selection.html

🔗 http://scikit-learn.org/stable/model_selection.html

Domingos, P. (2012). A few useful things to know about machine learning. *Communications of the ACM*, *55*(10), 78-87.

🔗 https://cacm.acm.org/magazines/2012/10/155531-a-few-useful-things-to-know-about-machine-learning/fulltext

(Challenge) Chapter 5, Section 1. Murphy, K. (2012). Machine Learning: A Probabilistic Perspective. Cambridge: The MIT Press.

🔗 https://probml.github.io/pml-book/book0.html

Session 55: Logistic regression

Learning Outcomes

#algorithms: Apply algorithmic thinking strategies to solve problems and effectively implement working code. (C) FA

Reading, Videos, and Other Preparation Resources

(Option 1) Read Sections 17.4 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

(Option 2) Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective.* Cambridge: The MIT Press.

🔗 https://probml.github.io/pml-book/book0.html

Session 70: Linear Algebra 3: Support Vector Machines MERGE WITH KERNEL TRICK

Learning Outcomes

#dataviz: Interpret, analyze, and create data visualizations. (C) EA

#regression: Apply and interpret regression. (C) FA

#cs156-MLCode: Produce working, readable, and performant Python implementations of a variety of machine learning systems using appropriate libraries and software tools.

#cs156-MLDevelopment: Contribute to the quality of ML resources for current and future students by compiling learning resources, sharing code, creating study groups, and supporting other student's learning.

#cs156-MLExplaination: Clearly articulate machine learning systems, algorithms, and techniques using appropriate oral and written descriptions, mathematical notation, and visualizations.

#cs156-MLFlexibility: Reason flexibly, apply information in new contexts, produce novel work, and articulate meta-knowledge about machine learning.

Reading, Videos, and Other Preparation Resources

Winston, P. (2014, January 10). Learning: Support Vector Machines. Retrieved September 01, 2017, from https://www.youtube.com/watch?v=_PwhiWxHK8o

https://www.youtube.com/watch?v=_PwhiWxHK8o

(Optional) Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective.* Cambridge: The MIT Press.

https://probml.github.io/pml-book/book0.html

Session 74: Kalman filters

Learning Outcomes

#modeling: Recognize how models can be used to describe a system, explain a set of data, and generate predictions. (C) EA

Reading, Videos, and Other Preparation Resources

Visually Explained Kalman Filters

https://www.youtube.com/watch?v=IFeCIbljreY

Labbe, R. (n.d.). Kalman and Bayesian Filters in Python. Retrieved October 30, 2017, from http://nbviewer.jupyter.org/github/rlabbe/Kalman-and-Bayesian-Filters-in-Python/blob/master/table_of_contents.ipynb

http://nbviewer.jupyter.org/github/rlabbe/Kalman-and-Bayesian-Filters-in-Python/blob/master/table_of_contents.ipynb

(Optional) Read Chapter 24 of Barber, D. (2012). *Bayesian reasoning and machine learning*. Cambridge: Cambridge University Press.

(Optional) Read Section 13.3 of Bishop, C. M. (2006). *Pattern recognition and machine learning*. New York: Springer.

Course Policies and Assessment

Please refer to the following pages on myMinerva regarding general course policies and assessment:

- https://my.minerva.edu/academics/policies/course-policies/
- https://my.minerva.edu/academics/policies/course-grading-assessment/

For the details regarding the overall course score calculation in this course, please refer to the “Course Scoring” section of this syllabus, above. The same information is reflected in the Outcome Index on Forum: from the "Courses" tab, click on "View Course Score Calculation".

Note: There may be minor changes to the syllabus before class launch due to integrating some new AI assessment tools and course structures. These are unlikely to substantially impact the content or class policies.

For this course see the syllabus on notion for more detail.